PROPERTY (V) IMPLIES THE COMPARISON OF HIGHER DIRECT IMAGES IN DEGREES UP TO N

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all degrees up to n, including the separate treatment of the top degree; no step is delegated or left to the reader.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The comparison situation of the section. Same chapter, Situation situation-compare and the standing setup of Section section-compare. Supplies epsilon, the subcategories A_X and A'_X, the subset P and the assumptions (1)-(5).
- The relative Leray spectral sequence. Same chapter, Lemma lemma-relative-Leray. Supplies the spectral sequence whose second page is analysed.
- Locality of cohomology and the vanishing of higher direct images. Same chapter, Lemmas lemma-kill-cohomology-class-on-covering and lemma-higher-direct-images. Supplies the vanishing of the tau-sheafification of the last term of the exact sequence.

Source reference. Original section 0EWK: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6177–6177. Navigation pointer invoked at proof line 6813; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EZH: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7425–7425. Navigation pointer invoked at proof

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line 6905; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EZG: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7361–7361. Navigation pointer invoked at proof line 6906; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 0EZ2: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6817–6817. Navigation pointer invoked at proof line 6900; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0734: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2367–2367. Navigation pointer invoked at proof line 6940; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01FW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 715–715. Navigation pointer invoked at proof line 7013; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 072W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 747–747. Navigation pointer invoked at proof line 7014; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. With $\mathcal{C}$, τ, and τ' as in Section 0EWK. Assume we are given a subset $\mathcal{P} \subset \text{Arrows}(\mathcal{C})$ and for every object X of $\mathcal{C}$ we are given a weak Serre subcategory $\mathcal{A}'_X \subset Ab(\mathcal{C}_{\tau'}/X)$. We make the following assumption:

(1) given $f : X \to Y$ in $\mathcal{P}$ and $Y' \to Y$ general, then $X \times_Y Y'$ exists and $X \times_Y Y' \to Y'$ is in $\mathcal{P}$,
(2) $f_{\tau'}^{-1}$ sends $\mathcal{A}'_Y$ into $\mathcal{A}'_X$ for any morphism $f : X \to Y$ of $\mathcal{C}$,
(3) given X in $\mathcal{C}$ and $\mathcal{F}'$ in $\mathcal{A}'_X$, then $\mathcal{F}'$ satisfies the sheaf condition for τ-coverings, i.e., $\mathcal{F}' = \epsilon_{X,*}\epsilon_X^{-1}\mathcal{F}'$,
(4) if $f : X \to Y$ in $\mathcal{P}$ and $\mathcal{F}' \in \text{Ob}(\mathcal{A}'_X)$, then $R^i f_{\tau',*}\mathcal{F}' \in \text{Ob}(\mathcal{A}'_Y)$ for $i \geq 0$.
(5) if $\{U_i \to U\}_{i\in I}$ is a τ-covering, then there exist
   (a) a τ'-covering $\{V_j \to U\}_{j\in J}$,
   (b) a τ-covering $\{f_j : W_j \to V_j\}$ consisting of a single $f_j \in \mathcal{P}$, and
   (c) a τ'-covering $\{W_{jk} \to W_j\}_{k\in K_j}$
   such that $\{W_{jk} \to U\}_{j\in J, k\in K_j}$ is a refinement of $\{U_i \to U\}_{i\in I}$.

Our next goal is to prove Lemmas 0EZH and 0EZG. We will do this by an induction argument using the following induction hypothesis.

(V_n) For X in $\mathcal{C}$ and $\mathcal{F}$ in $\mathcal{A}_X$ we have $R^i\epsilon_{X,*}\mathcal{F} = 0$ for $1 \leq i \leq n$.

Lemma 0.2. *In Situation 0.1 assume (V_n) holds. For $f : X \to Y$ in $\mathcal{P}$ and $\mathcal{F}$ in $\mathcal{A}_X$ we have $R^i f_{\tau',*}\epsilon_{X,*}\mathcal{F} = \epsilon_{Y,*}R^i f_{\tau,*}\mathcal{F}$ for $i \leq n$.*

Proof. We will use the commutative diagram (0EZ2) without further mention. In particular have

$$Rf_{\tau',*}R\epsilon_{X,*}\mathcal{F} = R\epsilon_{Y,*}Rf_{\tau,*}\mathcal{F}$$

Assumption (V_n) tells us that $\epsilon_{X,*}\mathcal{F} \to R\epsilon_{X,*}\mathcal{F}$ is an isomorphism in degrees $\leq n$. Hence $Rf_{\tau',*}\epsilon_{X,*}\mathcal{F} \to Rf_{\tau',*}R\epsilon_{X,*}\mathcal{F}$ is an isomorphism in degrees $\leq n$. We conclude that

$$R^if_{\tau',*}\epsilon_{X,*}\mathcal{F} \to H^i(R\epsilon_{Y,*}Rf_{\tau,*}\mathcal{F})$$

is an isomorphism for $i \leq n$. We will prove the lemma by looking at the second page of the spectral sequence of Lemma 0734 for $R\epsilon_{Y,*}Rf_{\tau,*}\mathcal{F}$. Here is a picture:

$$\begin{array}{cccc} \ldots & \ldots & \ldots & \ldots \\ \epsilon_{Y,*}R^2f_{\tau,*}\mathcal{F} & R^1\epsilon_{Y,*}R^2f_{\tau,*}\mathcal{F} & R^2\epsilon_{Y,*}R^2f_{\tau,*}\mathcal{F} & \ldots \\ \epsilon_{Y,*}R^1f_{\tau,*}\mathcal{F} & R^1\epsilon_{Y,*}R^1f_{\tau,*}\mathcal{F} & R^2\epsilon_{Y,*}R^1f_{\tau,*}\mathcal{F} & \ldots \\ \epsilon_{Y,*}f_{\tau,*}\mathcal{F} & R^1\epsilon_{Y,*}f_{\tau,*}\mathcal{F} & R^2\epsilon_{Y,*}f_{\tau,*}\mathcal{F} & \ldots \end{array}$$

Let (C_m) be the hypothesis: $R^if_{\tau',*}\epsilon_{X,*}\mathcal{F} = \epsilon_{Y,*}R^if_{\tau,*}\mathcal{F}$ for $i \leq m$. Observe that (C_0) holds. We will show that $(C_{m-1}) \Rightarrow (C_m)$ for $m < n$. Namely, if (C_{m-1}) holds, then for $n \geq p > 0$ and $q \leq m-1$ we have

$$\begin{aligned} R^p\epsilon_{Y,*}R^qf_{\tau,*}\mathcal{F} &= R^p\epsilon_{Y,*}\epsilon_Y^{-1}\epsilon_{Y,*}R^qf_{\tau,*}\mathcal{F} \\ &= R^p\epsilon_{Y,*}\epsilon_Y^{-1}R^qf_{\tau',*}\epsilon_{X,*}\mathcal{F} = 0 \end{aligned}$$

First equality as $\epsilon_Y^{-1}\epsilon_{Y,*} = \mathrm{id}$, the second by (C_{m-1}), and the final by by (V_n) because $\epsilon_Y^{-1}R^qf_{\tau',*}\epsilon_{X,*}\mathcal{F}$ is in $\mathcal{A}_Y$ by (4). Looking at the spectral sequence we see that $E_2^{0,m} = \epsilon_{Y,*}R^mf_{\tau,*}\mathcal{F}$ is the only nonzero term $E_2^{p,q}$ with $p+q=m$. Recall that $\mathrm{d}_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$. Hence there are no nonzero differentials $\mathrm{d}_r^{p,q}$, $r \geq 2$ either emanating or entering this spot. We conclude that $H^m(R\epsilon_{Y,*}Rf_{\tau,*}\mathcal{F}) = \epsilon_{Y,*}R^mf_{\tau,*}\mathcal{F}$ which implies (C_m) by the discussion above.

Finally, assume (C_{n-1}). The same analysis shows that $E_2^{0,n} = \epsilon_{Y,*}R^nf_{\tau,*}\mathcal{F}$ is the only nonzero term $E_2^{p,q}$ with $p+q=n$. We do still have no nonzero differentials entering this spot, but there can be a nonzero differential emanating it. Namely, the map $d_{n+1}^{0,n} : \epsilon_{Y,*}R^nf_{\tau,*}\mathcal{F} \to R^{n+1}\epsilon_{Y,*}f_{\tau,*}\mathcal{F}$. We conclude that there is an exact sequence

$$0 \to R^nf_{\tau',*}\epsilon_{X,*}\mathcal{F} \to \epsilon_{Y,*}R^nf_{\tau,*}\mathcal{F} \to R^{n+1}\epsilon_{Y,*}f_{\tau,*}\mathcal{F}$$

By (4) and (3) the sheaf $R^nf_{\tau',*}\epsilon_{X,*}\mathcal{F}$ satisfies the sheaf property for τ-coverings as does $\epsilon_{Y,*}R^nf_{\tau,*}\mathcal{F}$ (use the description of ϵ_* in Section 0EWK). However, the τ-sheafification of the τ'-sheaf $R^{n+1}\epsilon_{Y,*}f_{\tau,*}\mathcal{F}$ is zero (by locality of cohomology; use Lemmas 01FW and 072W). Thus $R^nf_{\tau',*}\epsilon_{X,*}\mathcal{F} \to \epsilon_{Y,*}R^nf_{\tau,*}\mathcal{F}$ has to be an isomorphism and the proof is complete. □